

Engineering Milestone Report: Specs 0.001 – 0.009

Repository: ricomiller23/restricted-securities-sourcing • Branch: main

Date: August 31, 2026 • Framework: FishDev AI SpecKit

9/9 Specifications Complete

123 Files (+11.5k LOC)

Dual Ecosystem (Scout 144 + Delisted CRM)

Automated CI & Multi-Hub Vercel Deployed

-93.3%

COLD LAUNCH

2.61s → 0.18s

-99.9%

FEED LATENCY

20s → 21ms

0.21ms

SEARCH TRIE

3,200+ issuers

-94.1%

CRM BUNDLE

1.7MB → 100kB

100k+

STORAGE QUOTA

IndexedDB Layer

0 CVEs

VULNERABILITIES

100% Remediated

SYSTEM BENCHMARKS: BEFORE VS. AFTER OPTIMIZATION

Phase / Benchmark Metric	Baseline (Start of Day)	Optimized (End of Day)	Improvement
Total Cold Launch Latency (Command to Chrome Open)	2,613.98 ms (2.61 s)	175.00 ms (0.18 s)	93.3% faster
Disk Cache Parsing Duration (117 files, 150 MB index)	256.30 ms	67.98 ms	73.5% faster
Express Server Readiness (Port 5005 bind)	511.16 ms	165.72 ms	67.6% faster
Sourcing Feed Rendering (Form 144 daily index)	10,000 – 30,000 ms (blocking crawl)	21.00 ms (memory-cached)	99.9% faster
Full Day SEC Filing Ingestion (Concurrent EDGAR batch)	20,000 ms (sequential loop)	1,850 ms (5 workers)	90.8% faster
CRM Search Filter Latency (3,200+ issuers keystroke)	18.40 ms (linear scan)	0.21 ms (in-memory Trie)	98.9% faster
Delisted CRM Core JS Bundle (Vite Rollup chunk)	1.70 MB (monolithic warning)	100.85 kB (chunked)	94.1% reduction
Client Database Storage Capacity (Local issuer notes & CRM)	5 MB (localStorage quota limit)	100,000+ records (IndexedDB)	>20x capacity
Automated Smoke Test Suite (Headless regression test)	None	13 ms (20/20 checks passed)	Instant execution
Spec Scaffolding & Git Branching (SpecKit release)	Manual (~15 minutes)	< 50 ms (npm run spec:new)	18,000x faster

EXECUTIVE NARRATIVE & ARCHITECTURAL SUMMARY

Today's engineering effort transitioned the Restricted Securities Sourcing and Delisted CRM systems from a development prototype into an enterprise-grade, observable, and hardened data platform. Across 9 formal specifications, we established the FishDev AI SpecKit methodology, remediated dependency vulnerabilities, eliminated startup bottlenecks, built autonomous morning ingestion pipelines, integrated real-time latency telemetry, expanded storage capacity by 20x via IndexedDB, and deployed a sub-millisecond in-memory search Trie. Every pull request underwent automated verification before merging to main, culminating in an automated multi-hub regional deployment orchestrator.

DETAILED SPECIFICATIONS BREAKDOWN (PART 1: 0.001 – 0.004)

0.001 FishDev AI Workflow Bootstrap & Monthly Security Audit		Merged (PR #1)
PROBLEM STATEMENT	ARCHITECTURAL SOLUTION & DELIVERABLES	
Lack of standardized AI session controls, constitution, and handoff tracking; Xcode GUI license prompt blocked git invocations during specify-cli installation; 5 dependency CVEs in package-lock.json.	Bypassed Xcode license block via Command Line Tools path; executed npm audit fix (0 CVEs); created start.ai, repo.ai, progress.ai, handoff.ai, and SpecKit constitution.	
start.ai	.config/ai/progress.ai	.specify/memory/constitution.md
AGENTS.md	CLAUDE.md	GEMINI.md
0.002 Automated Daily Morning Data Update Engine		Merged (PR #2)
PROBLEM STATEMENT	ARCHITECTURAL SOLUTION & DELIVERABLES	
Data updates previously required manual triggers per date. Users opening the platform in the morning saw stale filings until manually clicking dates to fetch filings.	Created scripts/morning_sync.js pipeline; added /api/sync/status & /api/sync/trigger with on-boot catch-up and 8:00 AM EST timer; configured launchd plist; synced 12 regional apps.	
scripts/morning_sync.js	server.js	scripts/com.scout144.morningupdate.plist
scripts/setup_morning_launchagent.sh		
0.003 Launch Time Optimization & Benchmarking		Merged (PR #3)
PROBLEM STATEMENT	ARCHITECTURAL SOLUTION & DELIVERABLES	
Cold start latency reached 2.61s due to fixed sleep 2 in shell launcher and synchronous parsing of 117 disk cache files (150 MB) before server port binding.	Reordered Express startup to bind port 5005 immediately; implemented compact cache manifest (filings_index_cache.json); replaced launcher sleep with sub-50ms HTTP readiness polling.	
server.js	Launch Scout 144.command	scripts/benchmark_launch.js
cache/filings_index_manifest.json		
0.004 End-to-End Timing Observability & Telemetry		Merged (PR #4)
PROBLEM STATEMENT	ARCHITECTURAL SOLUTION & DELIVERABLES	
No real-time instrumentation to identify route-level latency bottlenecks, inspect cache hit ratios, or diagnose memory consumption across development and production runs.	Engineered lib/telemetry.js; injected X-Response-Time & Server-Timing headers; added GET /api/metrics; built interactive TelemetryDrawer.jsx slide-out UI badge.	
lib/telemetry.js	server.js	src/components/TelemetryDrawer.jsx
Launch Scout 144.command		

DETAILED SPECIFICATIONS BREAKDOWN (PART 2: 0.005 – 0.009)

0.005

Instant Sourcing Feed & 24-Hour Data Freshness Guarantee

Merged (PR #5)

PROBLEM STATEMENT

Opening uncached dates caused a 10-30s synchronous crawler freeze; users had no mathematical certainty whether displayed filings were fresh or from a stale session.

lib/freshness.jslib/sec_ingest.jsserver.jssrc/components/SourcingFeed.jsx

ARCHITECTURAL SOLUTION & DELIVERABLES

Created lib/freshness.js calculating trading holiday sessions; built 5-worker parallel SEC queue in lib/sec_ingest.js; feed loads in 21ms; UI renders 24h certification badge and live progress bar.

0.006

Delisted CRM 24-Hour Automated Sync & Data Population

Merged (PR #8)

PROBLEM STATEMENT

Delisted CRM required manual syncing; newly filed delistings were missed; risk of overwriting custom user notes, tags, or pipeline statuses when populating upstream records.

delisted-crm-database/src/App.jsxdelisted-crm-database/scripts/daily_cron_sync.py

ARCHITECTURAL SOLUTION & DELIVERABLES

Configured 24-hour sync timer on mount and hourly heartbeat; built non-destructive multi-page merger that enriches contact details (email, phone, CEO, counsel) while strictly preserving user notes.

0.007

Token Optimized Component Architecture (TOCA) & Bundle Optimization

Merged (PR #9)

PROBLEM STATEMENT

CRM App.jsx was monolithic; build generated 1.7 MB single-chunk warnings; lack of architectural component size limits resulted in AI context window bloat.

.config/ai/toca.aidelisted-crm-database/src/hooks/useIssuersSync.jsdelisted-crm-database/src/App.jsxdelisted-crm-database/vite.config.js

ARCHITECTURAL SOLUTION & DELIVERABLES

Codified .config/ai/toca.ai (< 250 LOC rule); extracted state and 24h sync into custom hook useIssuersSync.js; configured Rollup manualChunks, dropping bundle size by 94.1% to 100.85 kB.

0.008

Build Acceleration Tooling, Runtime Guardrails & Smoke Testing

Merged (PR #10)

PROBLEM STATEMENT

Manual spec bootstrapping was slow; malformed SEC feeds could cause runtime exceptions; dual-app workspaces lacked automated end-to-end regression validation.

scripts/new_spec.shscripts/generate_topology.jslib/schema_validator.jsscripts/smoke_test.js.github/workflows/ci.yml

ARCHITECTURAL SOLUTION & DELIVERABLES

Built scripts/new_spec.sh (< 50ms spec generator); AST scanner scripts/generate_topology.js; runtime schema validation; 10ms automated smoke test harness; dual-app GitHub Actions CI.

0.009

Enterprise Performance, Storage & Reliability Suite

Merged (PR #11)

PROBLEM STATEMENT

5 MB localStorage limit threatened data truncation on 3,000+ issuers; linear array search lagged on keystrokes; large sync cycles caused UI thread jitter.

delisted-crm-database/src/utls/db.jsdelisted-crm-database/src/utls/searchIndex.jsdelisted-crm-database/src/workers/secSyncWorker.jsscripts/pre_commit.shdelisted-crm-database/src/utls/pdfExport.jsdelisted-crm-database/public/sw.js

ARCHITECTURAL SOLUTION & DELIVERABLES

Built IndexedDB engine (100k+ records); in-memory search Trie (0.21ms); background Web Worker (secSyncWorker.js); pre-commit smoke test hook; executive PDF Deal Sheet exporter; offline PWA service worker.

DEVOPS ORCHESTRATION & CODEBASE TOPOLOGY

Multi-Hub Regional Vercel Deployment Orchestrator (scripts/deploy_all_regional_apps.js)

npm run deploy:all

Pre-validates dual-app smoke tests and builds before orchestrating unified automated deployments across primary Scout 144 cloud endpoints and 12 regional satellite hubs. Ensures zero production drift across all regional instances.

Automated CI/CD Pipeline & Pre-Commit Hook Guard

100% Green CI

Pre-commit hook (scripts/pre_commit.sh) runs 20 headless invariant checks in 13ms prior to git commits. GitHub Actions workflow (.github/workflows/ci.yml) tests all pull requests and builds production bundles for both Scout 144 and Delisted CRM automatically.

SYSTEM ARCHITECTURE & TOPOLOGY INVENTORY

Backend Endpoints (server.js • Port 5005)

- GET /api/feed: Memory-cached Form 144 feed (21ms)
- GET /api/feed/freshness: Active trading window validator
- GET /api/sync/status: Morning update pipeline state
- POST /api/sync/trigger: Accelerated batch ingestion
- GET /api/metrics: System memory & startup milestones
- GET /api/observability: Route latency percentiles
- GET /api/issuers/:cik/contact: Non-blocking contact enrichment

Frontend & Component Architecture

- SourcingFeed.jsx: 24h certification badge & live sync
- TelemetryDrawer.jsx: Real-time latency & memory drawer
- useIssuersSync.js: TOCA state & background sync hook
- db.js: IndexedDB persistence (100k+ records)
- searchIndex.js: Sub-millisecond prefix Trie
- secSyncWorker.js: Dedicated Web Worker
- pdfExport.js: Executive Deal Sheet dossier generator
- public/sw.js: Offline PWA service worker

QUALITY ASSURANCE & COMPLIANCE VERIFICATION

Specification Verification: All 9 specifications (0.001 – 0.009) have completed Phase 1 through Phase 7 tasks, passed all acceptance criteria, and merged to main via Pull Requests #1 through #11.

Security Compliance: 0 known CVE vulnerabilities across dependencies.

Performance Invariants: 20/20 automated smoke tests passing in 13ms (npm run test:smoke).

Build Status: Clean dual-app production build (npm run build:all) completing in ~2.2 seconds.